THE COPPERBELT UNIVERSITY

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

SOFTWARE REQUIREMENTS SPECIFICATION

PROJECT TITLE: THE KEY PERFORMANCE INDICATOR (KPI) EVALUATION SYSTEM

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1. INTRODUCTION

This document describes the software requirements for the KPI Evaluation System for the School of Information and Communication Technology (SICT) at The Copperbelt University. The system is designed to assist in managing, tracking, and evaluating academic staff performance using defined Key Performance Indicators (KPIs) in real time.

2. USER NEEDS

- Faculty and staff need a personalized dashboard to track performance metrics.
- Supervisors require automated evaluation tools to assess employee performance objectively.
- University administrators need data-driven insights for decision-making.
- The system should support real-time data visualization (graphs, charts, reports).
- Users must have secure access with role-based permissions.

3. FUNCTIONAL REQUIREMENTS

User Authentication & Role Management

- Secure login with role-based access control (faculty, department heads, and administrators).
- Define KPI: the system must allow users to define new KPI including name, description, weight target value and units of measure.
- Setting targets the system must allow users to set targets for each KPI, as well as define thresholds to trigger alerts or highlight performance issues.
- KPI assignment: Administrator should be able to assign specified KPI to individual lectures or groups of lectures based on their roles and responsibilities, and update KPIs for faculty and staff.
- KPI review and modification: administrator should be able to review, edit and delete existing KPIs.
- Data input and collection: the system should allow lectures to submit data related to their assigned KPI (e.g. work plans).
- Automatically calculate KPI scores based on predefined criteria: the system should automatically perform the calculations on predefined criteria and collected data.
- Reporting and visualizing: the system should provide visual representation if lecture performance against targets.
- Different views should be available for individual lectures like HOD and administrator.

Performance Tracking & Evaluation

- Record faculty contributions (teaching, research, administration).
- Generate automated performance reports for supervisors.
- Compare performance trends over time.

Data Visualization & Reporting

- Reports should be exported in various formats like PDF, CSV.
- Administrator should be able to customize report parameters like the data range, department, and lectures.
- Provide interactive dashboards for real-time performance monitoring.
- Allow users to download KPI reports in PDF/Excel format.
- Notifications & Alerts.
- Send alerts for missed targets, deadlines, and evaluations.

Audit & Logging System

- Maintain activity logs for transparency in evaluations.

4. NON-FUNCTIONAL REQUIREMENTS

- **Security**: Implement encryption and compliance with data protection policies.
- **Responsiveness**: the system should be responsive and load pages quickly for all users, actions like data generation and navigation and data submission should be swift.
- **Scalability**: System should support future expansion for additional functionality.
- **Efficiency**: the system should utilize resources like server processing, memory efficiently to minimal cost and optimal performance.
- **Integrity**: the system must ensure the accuracy and completeness of the data. Measures should be put in place to prevent unauthorized modification or deletion of data.
- **Accessibility**: the system should be highly available and accessible to users with minimal downtime.
- **Ease of use**: the system should have a clear, interactive and user-friendly interface that is easy to navigate and understand for all user roles.
- **Maintainability**: the system should follow modular design principles making it easier to understand, modify and update specific components without affecting the entire system.

5. CONSTRAINTS

- The system must adhere to the university security policies this include password complexity and regular security auditing.
- Timeline: there might be a specific deadline for the system to be developed and deployed.
- Internet Dependency: System requires stable internet connectivity.
- Data Accuracy: Performance evaluations rely on accurate data inputs.
- Resource Availability: Limited budget for implementation and maintenance.

6. PERFORMANCE REQUIREMENTS

Login:

- Upon a successful login, the user is redirected to their personal dashboard with relevant information and functionality based on their role.

KPI management:

- When an administrator creates a new KPI, it becomes available for assignment to lectures.
- Modification of KPIs should potentially have the changes logged and the relevant administrators should be notified.
- Deleting a KPI should prevent it from being assigned in future, but it must remain historical data.

Data input:

- The system must validate data input to ensure it conforms to expected format and range of each KPI. Aggregated feedback should only be made available to lectures and HODs after the feedback period closes.
- The system must automatically trigger KPI calculations based on defined schedules or events, calculated scores should be reflected in reports. If performance metrics fall below expectations, the system triggers alerts.
- When reporting, generated reports should reflect accurate and correct data.
- Users can generate and download custom performance reports.

7. SOFTWARE AND HARDWARE REQUIREMENTS

7.1. HARDWARE REQUIREMENTS:

Development Environment Hardware

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 8 GB
- **Storage:** HDD with minimum 100 GB free space.
- **Graphics:** Standard integrated graphics (no GPU dependency).
- **Display:** Minimum 720p resolution.

Deployment Server Hardware (Web Server)

- **Processor:** Intel Xeon or equivalent (4-core minimum)
- **RAM:** 16 GB minimum
- **Storage:** 500 GB SSD
- **Network:** Gigabit Ethernet or high-speed internet connectivity.
- **Backup:** External hard drive or cloud backup system, for weekly snapshots.

7.2. SOFTWARE REQUIREMENTS:

Operating System

- **Development OS:** Windows 10 or Windows 11
- **Server OS:** Windows Server 2019

Backend and Compiler Tools

- **Programming Language:** C# (ASP.NET Core framework)
- **Compiler/IDE:** Visual Studio Code or Visual Studio 2022
- **Package Manager:** NuGet (for C# dependencies)

Frontend Tools

- **Language:** JavaScript (React.js)
- **Libraries/Frameworks:**
    - React.js
    - Bootstrap 5
    - Recharts (for chart visualizations).
- **Package Manager:** npm (Node.js must be installed).

Database Tools

- **DBMS:** MySQL 8.0 or later.
- **Tool:** MySQL Workbench for database modeling and management.

Testing and Version Control Tools

- **Testing Tools:** Jest (for frontend), Postman (for API testing).
- **Version Control:** Git (GitHub for collaboration and repository management).

System Development Platform

- **Frontend Platform:** React.js with Bootstrap and CSS modules.
- **Backend Platform:** ASP.NET Core (C#)
- **Database:** MySQL
- **IDE & Editors:** Visual Studio Code, Visual Studio 2022.
- **Browser Compatibility:** Google Chrome, Mozilla Firefox, Microsoft Edge.
- **Hosting (optional):** Can be deployed locally or hosted on a university cloud/web server.

8. PRELIMINARY PRODUCT DESCRIPTION

The KPI Evaluation System for CBU-SICT is a web-based platform that enables academic supervisors to define, assign, and evaluate key performance indicators for staff. The system will support the generation of performance reports while offering export functionality. It allows users at different administrative levels (Lecturer, HOD, Dean) to access performance data relevant to their roles.

Functional Overview

- Secure login for different user roles
- KPI assignment per academic year or term
- Dashboard views with performance graphs
- Evaluation based on predefined rating scales
- Export and print options for reports.

9. ORGANIZATIONAL REQUIREMENTS

Operational Requirements

- The system must be accessible by authenticated users within the university network.
- Role-based access should restrict views and actions based on user designation (e.g., Lecturer, Dean Etc…).

Development Requirements

- The system must be developed using ASP.NET (C#) for backend and React.js for frontend.
- Code must be modular and support future feature extensions.
- GitHub will be used for collaborative control.

Environmental Requirements

- The system should operate in a web browser on both desktop and mobile devices.
- It should be deployable on a Windows-based web server with internet access.

10. EXTERNAL REQUIREMENTS

Requirements

- The system should adhere to CBU’s internal data access policies and any applicable national educational IT regulations.

Legislative Requirements

- The system must ensure user data privacy and must not expose personal evaluation results to unauthorized parties.
- The system should comply with Zambia’s Data Protection Act.

Ethical Requirements

- The system should provide a fair and unbiased evaluation framework for all users.
- KPI metrics and evaluations must be transparent and accessible for potential improvement purposes.